

Homework 7.3

Problem 19

Which of these polynomials is orthogonal to Π_2 on the interval $[0,1]$ with weight function $w(x) = 1$?

Now, to conclude that $p_n(x) \perp \Pi_2$, we require that $\forall f \in \Pi_2, \langle p_n, f \rangle_{w(x)} = 0$.

$$f(x) = ax^2 + bx + c, a, b, c \in \mathbb{R}$$

$$p_n \perp \Pi_2 \Leftrightarrow \langle p_n, f \rangle_{w(x)} = \int_0^1 p_n(x)f(x)w(x)dx = 0, \forall a, b, c \in \mathbb{R}$$

$$p_1(x) = 1 + x$$

$$\int_0^1 (1+x)(ax^2 + bx + c)dx \Rightarrow a \int_0^1 (x^2 + x^3)dx = a \left(\frac{1}{3} + \frac{1}{4} \right) \neq 0, \forall a \in \mathbb{R}$$

$$\therefore p_1 \text{ not } \perp \Pi_2$$

$$p_2(x) = x - \frac{1}{2}$$

$$p_1 \text{ not } \perp \Pi_2 \Rightarrow p_2 \text{ not } \perp \Pi_2$$

$$\therefore p_2 \text{ not } \perp \Pi_2$$

$$p_3(x) = x^2 - 3x + 1$$

$$\int_0^1 (x^2 - 3x + 1)(ax^2 + bx + c)dx$$

$$\Rightarrow a \int_0^1 (x^4 - 3x^3 + x^2)dx = a \left(\frac{1}{5} - \frac{3}{4} + \frac{1}{3} \right) = a \left(-\frac{13}{60} \right) \neq 0 \forall a \in \mathbb{R}$$

$$\therefore p_3 \text{ not } \perp \Pi_2$$

$$p_4(x) = 35x^4 - 60x^2 + 32x - 3$$

$$\int_0^1 (35x^4 - 60x^2 + 32x - 3)(ax^2 + bx + c)dx$$

$$a \int_0^1 (35x^6 - 60x^4 + 32x^3 - 3x^2)dx = a \left(\frac{35}{7} - \frac{60}{5} + \frac{32}{4} - \frac{3}{3} \right) = a(5 - 12 + 8 - 1) = 0$$

$$b \int_0^1 (35x^5 - 60x^3 + 32x^2 - 3x)dx = b \left(\frac{35}{6} - \frac{60}{4} + \frac{32}{3} - \frac{3}{2} \right) = b(0) = 0$$

$$c \int_0^1 (35x^4 - 60x^2 + 32x - 3)dx = c \left(\frac{35}{5} - \frac{60}{3} + \frac{32}{2} - 3 \right) = c(7 - 20 + 16 - 3) = 0$$

$$\therefore p_4 \perp \Pi_2$$

$$p_5(x) = x^3 - 3x^2 + x - 1$$

$$\int_0^1 (x^3 - 3x^2 + x - 1)(ax^2 + bx + c)dx$$

$$a \int_0^1 (x^5 - 3x^4 + x^3 - x^2)dx = a \left(\frac{1}{6} - \frac{3}{5} + \frac{1}{4} - \frac{1}{3} \right) = a \left(-\frac{31}{60} \right) \neq 0 \forall a$$

$$\therefore p_5 \text{ not } \perp \Pi_2$$

Computer Homework 7.5

Problem 1

Program the adaptive algorithm and test it on these integrals.

```
function Adaptive = Ad(f,a,b,ep,n)
% This program uses the algorithm given in the book to calculate the
% integral of a given function given certain parameters.

% This is going to be used to calculate the step size.
Del = b - a;
h = Del / 2;

% This will hold the value of the integral as we loop through our value.
I = 0;
% Value needed for Simpson's rule.
c = (a + b)/2;

% Matrix that will hold our stacks.
M = [];

% Function evaluations.
fa = f(a);
fb = f(b);
fc = f(c);

% Simpson's Rule.
S = (fa + 4*fc + fb)*h/3;

% Base vector and adding vector to stack.
V = [a,h,fa,fc,fb,S];
M = [M;V];

% Initialize our value for our loop.
k = 1;

while ((1 <= k) && (k <= n))

    % Doing Simpson's rule for the smaller interval.
    h = M(k,2)/2;
```

```

fy = f(M(k,1) + h);
SS = (M(k,3) + 4*fy + M(k,4))*h/3;
fz = f(M(k,1) + 3*h);
SSS = (M(k,4) + 4*fz + M(k,5))*h/3;

% This test whether we are withing our tolerance for our integral.
if (abs(SS + SSS - M(k,6)) < ((30*ep*h)/Del))
    I = I + SS + SSS + (SS + SSS - M(k,6))/15;
    k = k - 1;

    % If we are done
    if (k <= 0)
        Adaptive = I;
        break;
    end

else
    % This is to avoid leaving our loop and diverging.
    if k >= n
        'Failure'
        break;
    end

    % Redefining our vector for the adaptive.
    vv = M(k,5);

    M(k,2) = h;
    M(k,5) = M(k,4);
    M(k,4) = fy;
    M(k,6) = SS;
    M = [M;V];

    % Adding another vector to the stack.
    k = k + 1;
    VV = [M(k - 1,1)+2*h, h, M(k - 1,5),fz,vv,SSS];
    M = [M;VV];

end
end

```

a) $\int_0^1 x^{\frac{1}{2}} dx$

$$\int_0^1 x^{\frac{1}{2}} dx = \left[\frac{2x^{\frac{3}{2}}}{\frac{3}{2}} \right]_0^1 = \frac{2}{3}$$

```

>> f = @(x) x^(1/2);
>> Ad(f,0,1,1,10)
ans = 0.6578

```

b) $\int_0^1 (1-x)^{\frac{1}{2}} dx$

$$\int_0^1 (1-x)^{\frac{1}{2}} dx = \int_1^0 u^{\frac{1}{2}} (-du) = \frac{2}{3}$$

```
>> f = @(x) (1-x)^(1/2);
```

```
>> Ad(f,0,1,1,10)
```

```
ans = 0.6578
```

Homework 8.1

Problem 3

Show that $x = -t^2/4$ and $x = 1 - t$ are solutions of the initial-value problem

$$\begin{cases} 2x' = \sqrt{t^2 + 4x} - t \\ x(2) = -1 \end{cases}$$

$$x = -\frac{t^2}{4} \Rightarrow x' = -\frac{t}{2} \Rightarrow 2x' = -t \Rightarrow \sqrt{t^2 + 4x} - t = \sqrt{t^2 - t^2} - t = -t$$

$$\therefore 2x' = \sqrt{t^2 + 4x} - t, x(2) = -\frac{(2)^2}{4} = -1$$

$$x = 1 - t \Rightarrow x' = -1 \Rightarrow 2x' = -2$$

$$\Rightarrow \sqrt{t^2 + 4x} - t = \sqrt{t^2 + 4(1-t)} - t = \sqrt{t^2 - 4t + 4} - t = \sqrt{(t-2)^2} - t = t - 2 - t = -2$$

$$\therefore 2x' = \sqrt{t^2 + 4x} - t, x(2) = 1 - 2 = -1$$

Why does this not contradict Theorem 2, on initial-value problem uniqueness?

$$2f(x, t) = \sqrt{t^2 + 4x} - t \Rightarrow 2f_x = \frac{1}{2} (t^2 + 4x)^{-\frac{1}{2}} * 4 \Rightarrow f_x = (t^2 + 4x)^{-\frac{1}{2}}$$

$$(t, x) = (2, -1) \Rightarrow f_x: \text{not exists}$$

Thus, we do not satisfy the conditions of theorem 2.

Problem 5

Solve the initial-value problem $x' = f(x, t), x(0) = 0$ in the following cases. Use the fact that $\frac{dt}{dx} = (dx, dt)^{-1}$ when $\frac{dx}{dt} \neq 0$.

a) $f(x, t) = x^{-2}$

$$x' = x^{-2} \Rightarrow \frac{dx}{dt} = \frac{1}{x^2} \Rightarrow x^2 dx = dt \Rightarrow \frac{x^3}{3} + c = t$$

$$\frac{0}{3} + c = 0 \Rightarrow c = 0$$

$$\therefore x = \sqrt[3]{3t}$$

b) $f(x, t) = 1 + x^2$

$$x' = 1 + x^2 \Rightarrow \frac{dx}{1 + x^2} = dt \Rightarrow \arctan(x) = t + c$$

$$\arctan(0) = c \Rightarrow c = 0$$

$$\therefore x = \tan(t)$$

Problem 7

Show by using theorem 1, on initial-value problem existence, that the initial-value problem

$$\begin{cases} x' = \tan(x) \\ x(0) = 0 \end{cases}$$

Has a solution in the interval $|t| < \frac{\pi}{4}$.

The center of our rectangle will be $(0,0)$, $-\frac{\pi}{4} < t < \frac{\pi}{4}$

$$R = \{(t, x) : |t| \leq \alpha, |x| < \beta\}$$

$$f(x, t) = \tan(x)$$

We want $|t| \leq \frac{\pi}{4} = \min\left(\alpha, \frac{\beta}{M}\right)$, $M = \max_{(x,t) \in R} |f(x, t)|$. Since we know that $|\tan(x)| \rightarrow \infty$ as $x \rightarrow \pm \frac{\pi}{2}$, but since we want $\min\left(\alpha, \frac{\beta}{M}\right) = \frac{\pi}{4}$, we will need to make sure that $\frac{\beta}{M} \geq \frac{\pi}{4}$ so that $\alpha = \frac{\pi}{4}$ as we need. Since on the interval $\left(0, \frac{\pi}{2}\right)$, $\tan(x)$ is increasing, we know that $\max_{(x,t) \in R} |f(x, t)| = \tan(\beta)$

$$\frac{\beta}{\tan(\beta)} \geq \frac{\pi}{4} \Rightarrow \frac{\beta}{\tan(\beta)} \geq \frac{\pi}{4} \Rightarrow \beta \leq \frac{\pi}{4}$$

$$\therefore R = \left\{(t, x) : |t| \leq \frac{\pi}{4}, |x| < \frac{\pi}{4}\right\}$$

Homework 8.2

Problem 2

Verify that the function $x(t) = \frac{t^2}{4}$ solves the initial-value problem

$$\begin{cases} x' = \sqrt{x} \\ x(0) = 0 \end{cases}$$

$$x = \frac{t^2}{4} \Rightarrow x' = \frac{t}{2} = \sqrt{\frac{t^2}{4}} = \sqrt{x}$$

$$x(0) = \frac{0^2}{4} = 0$$

Apply the Taylor series method of order 1, and explain why the numerical solution differs from the solution $\frac{t^2}{4}$.

$$x_{i+1} = x_i + hf(x_i, t)$$

$$f(x, t) = \sqrt{x}, h = .25$$

t	0	.25	.50	.75	1.0
x_A : approximate	0	0	0	0	0
x_T : True	0	$\frac{1}{64}$	$\frac{1}{16}$	$\frac{9}{64}$	$\frac{1}{4}$

We have a solution since $f(x, t)$ is continuous for $x \geq 0$, but since we have that $f_x = \frac{1}{2\sqrt{x}}$, which is undefined for $(x, t) = (0, 0)$ we fail the uniqueness theorem and thus $x = 0$ is the solution we get although $x = \frac{t^2}{4}$ is also a solution.

Problem 5

Consider the ordinary differential equation

$$\begin{cases} 5tx' + x^2 = 2 \\ x(4) = 1 \end{cases}$$

Calculate $x(4.1)$ using the Taylor-series method of order 2.

$$x(t+h) = x(t) + hx'(t) + \frac{h^2}{2}x''(t) + \mathcal{O}(h^3)$$

$$x'(t) = \frac{2-x^2}{5t} \Rightarrow \frac{d}{dt}x' = \frac{5t(-2x * x') - (2-x^2)5}{(5t)^2}$$

$$t = 4, h = .1, x(4) = 1$$

$$x'(4) = \frac{2-x(4)^2}{5(4)} = \frac{2-1^2}{20} = \frac{1}{20}$$

$$x''(4) = \frac{5(4)(-2x(4) * x'(4)) - (2-x(4)^2)5}{(5(4))^2} = \frac{20 * -2 * \frac{1}{20} - 5}{20^2} = -\frac{7}{400}$$

$$x(4+.1) = x(4) + (.1)x'(4) + \frac{(.1)^2}{2}x''(4) = 1 + \frac{1}{10} * \frac{1}{20} + \frac{1}{200} * \left(-\frac{7}{400}\right) = 1 + \frac{1}{200} - \frac{7}{80000}$$

$$\therefore x(4.1) \approx 1.0049125$$

Problem 7

If the Taylor-series method is used to solve an initial-value problem involving the differential equation

$$x' = \cos(tx)$$

What are the formulas for x'' , x''' , and $x^{(4)}$?

$$x'' = \frac{d}{dt} x' = \frac{d}{dt} \cos(tx) = -\sin(tx) (x + tx')$$

$$\therefore x'' = -\sin(tx) (x + t \cos(tx))$$

$$x''' = \frac{d}{dt} (x'') = \frac{d}{dt} (-\sin(tx) * (x + tx')) = -\cos(tx) (x + tx')^2 - \sin(tx) (x' + x' + tx'')$$

$$\therefore x''' = -\cos(tx) (x + t \cos(tx))^2 - \sin(tx) (2 \cos(tx) - t \sin(tx) (x + t \cos(tx)))$$

$$x^{(4)} = \frac{d}{dt} x''' = \frac{d}{dt} (-\cos(tx) (x + tx')^2 - \sin(tx) (2x' + tx''))$$

$$\frac{d}{dt} (-\cos(tx) (x + tx')^2) = \sin(tx) (x + tx') (x + tx')^2 - \cos(tx) (2(x + tx') * (x' + x' + tx''))$$

$$\Rightarrow \frac{d}{dt} (-\cos(tx) (x + tx')^2) = \sin(tx) (x + tx')^3 - 2 \cos(tx) (x + tx') (2x' + tx'')$$

$$\frac{d}{dt} (-\sin(tx) (2x' + tx'')) = -\cos(tx) (x + tx') - \sin(tx) (2x'' + x'' + tx''')$$

$$\Rightarrow \frac{d}{dt} (-\sin(tx) (2x' + tx'')) = -\cos(tx) (x + tx') - \sin(tx) (3x'' + tx''')$$

$$\Rightarrow \frac{d}{dt} x''' = \sin(tx) (x + tx')^3 - 2 \cos(tx) (x + tx') (2x' + tx'') - \cos(tx) (x + tx') - \sin(tx) (3x'' + tx''')$$

$$\begin{aligned} \therefore x^{(4)} &= \sin(tx) (x + t \cos(tx))^3 \\ &\quad - 2 \cos(tx) (x + t \cos(tx)) (2 \cos(tx) + t(-\sin(tx) (x + t \cos(tx)))) \\ &\quad - \cos(tx) (x + t \cos(tx)) \\ &\quad - \sin(tx) (3(-\sin(tx) (x + t \cos(tx))) + t(-\cos(tx) (x + t \cos(tx))^2 \\ &\quad - \sin(tx) (2 \cos(tx) - t \sin(tx) (x + t \cos(tx)))) \end{aligned}$$